next digital

ORQUESTANDO TAREAS CON AIRFLOW: TUS DATOS EN PILOTO AUTOMÁTICO



Sesión 1: Introducción a Apache Airflow

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¿De dónde vengo?



Javier Sepulveda

- Grado+Master Ingenieria Civil UPM
- Desarrollador fullstack
- Ingeniero de plataforma

+140 especialistas

ÁREAS

- Cloud
- Data & AI
- Analítica Avanzada
- Desarrollo de Software

SECTORES













Azure

amazon web services

dbt

SirviendoCódigo.com

www.youtube.com/@SirviendoCódigo



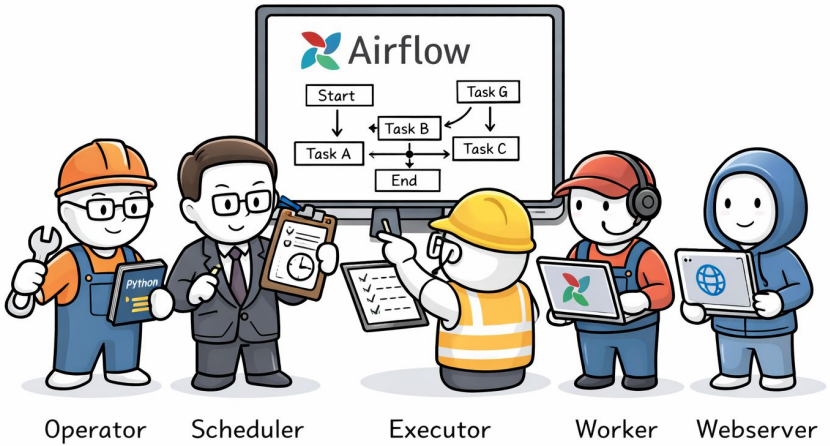
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2. Servicios de integración
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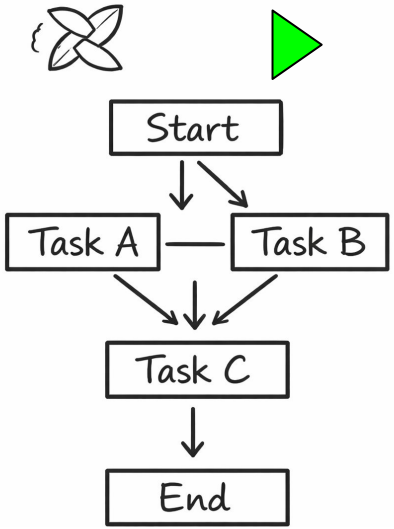


Objetivos

Entender qué es Airflow y sus componentes principales



Crear y ejecutar un DAG sencillo desde la interfaz de Airflow

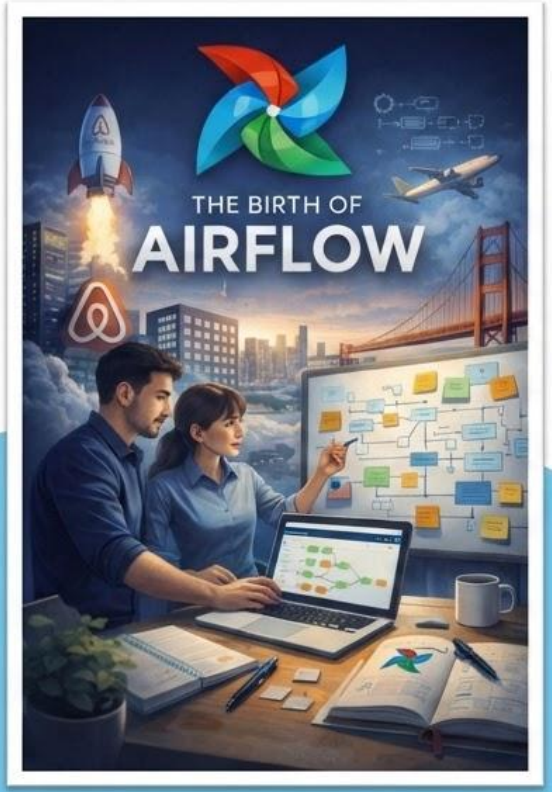




01_
¿Qué es Airflow?


Airflow
Extract
Transform
Load
XCom

Historia de Airflow



- Desarrollada como herramienta interna de Airbnb
- Donado como proyecto open source a la Apache Foundation bajo el nombre de Apache Airflow
- En funcionamiento desde octubre de 2014
- Objetivo: gestionar los ETLs de la compañía, cada vez más complejos
- En ese momento las herramientas no eran escalables ni fáciles de mantener

¿Qué es Airflow?



Plataforma *open-source* de gestión de flujos de trabajo



Automatizar, programar y monitorizar



Escalable, dinámico, extensible

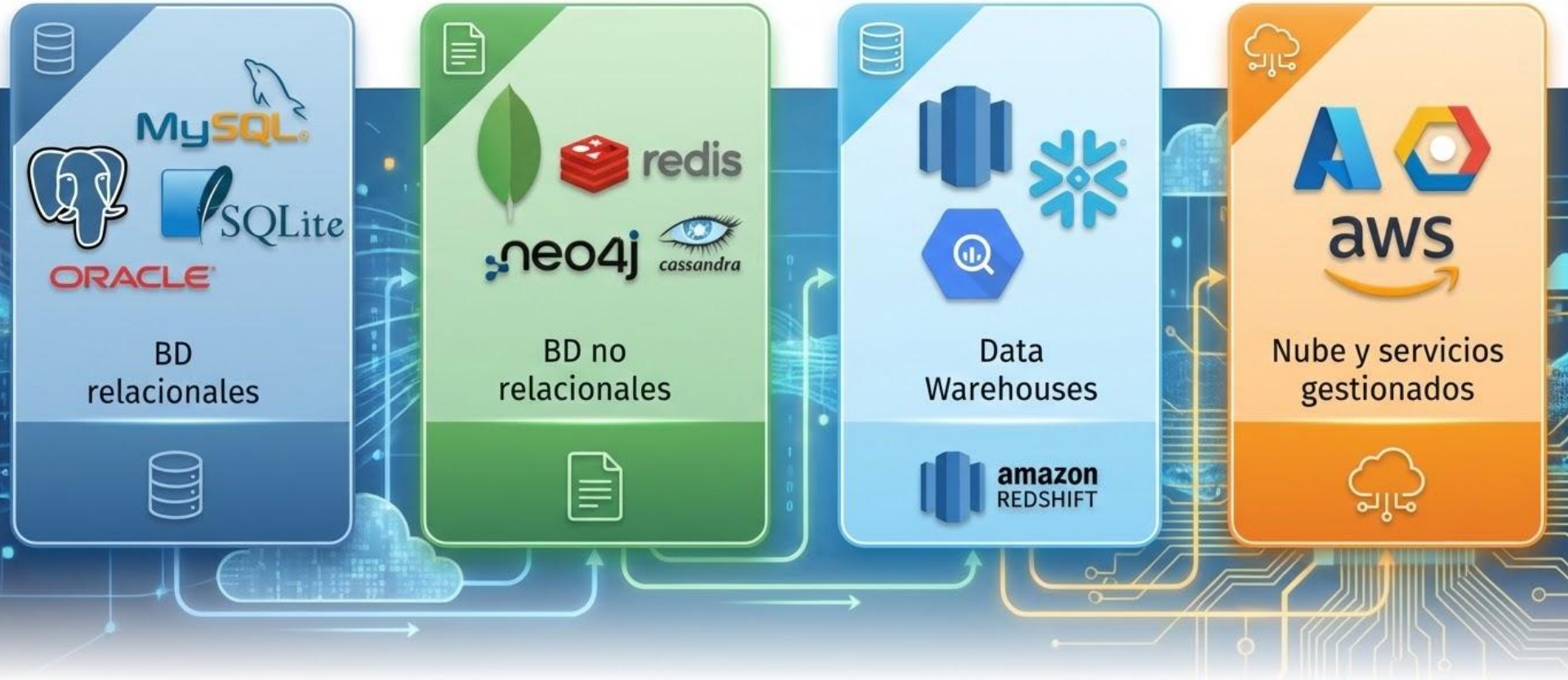


Escrito en Python



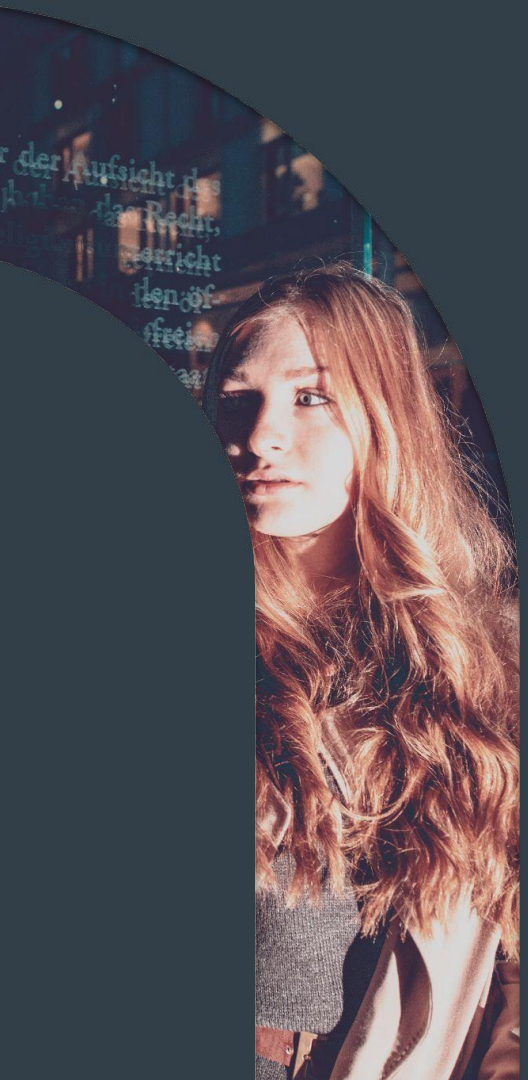
02_
Servicios de integración

Servicios de integración



Servicios de integración

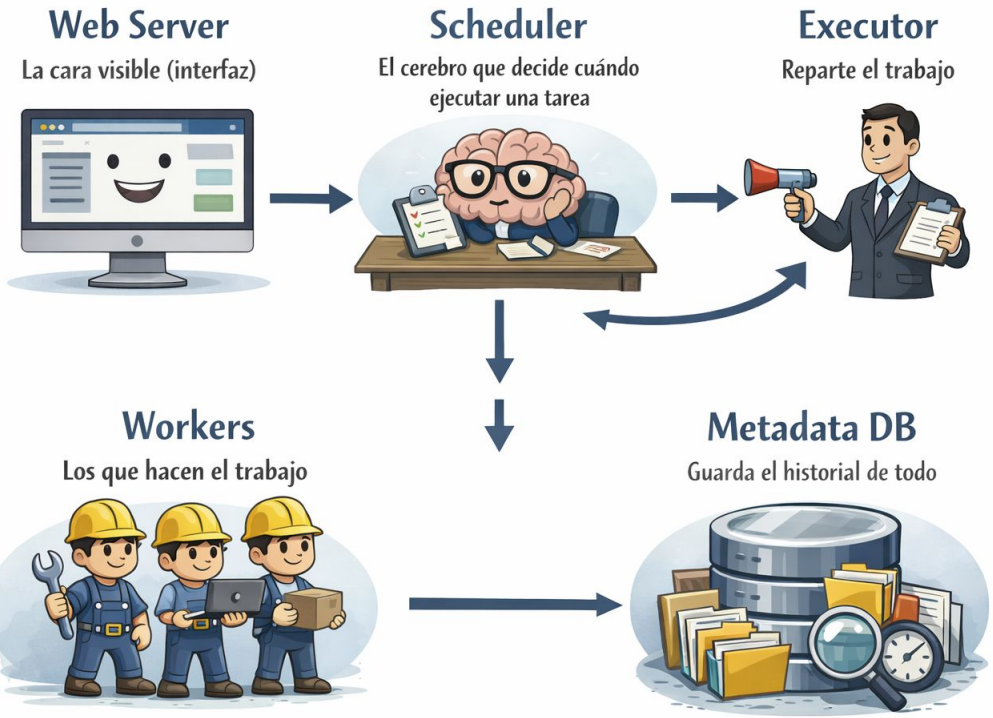




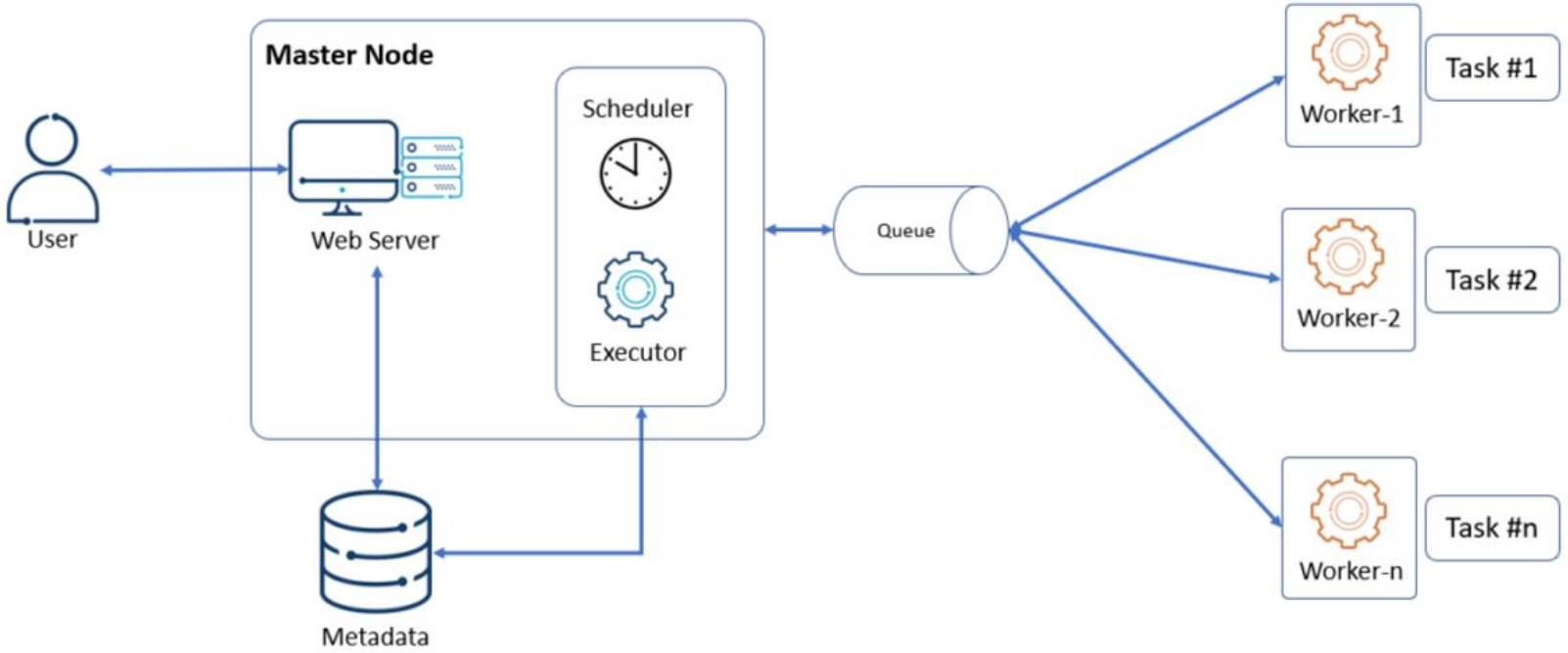
03_

Componentes principales

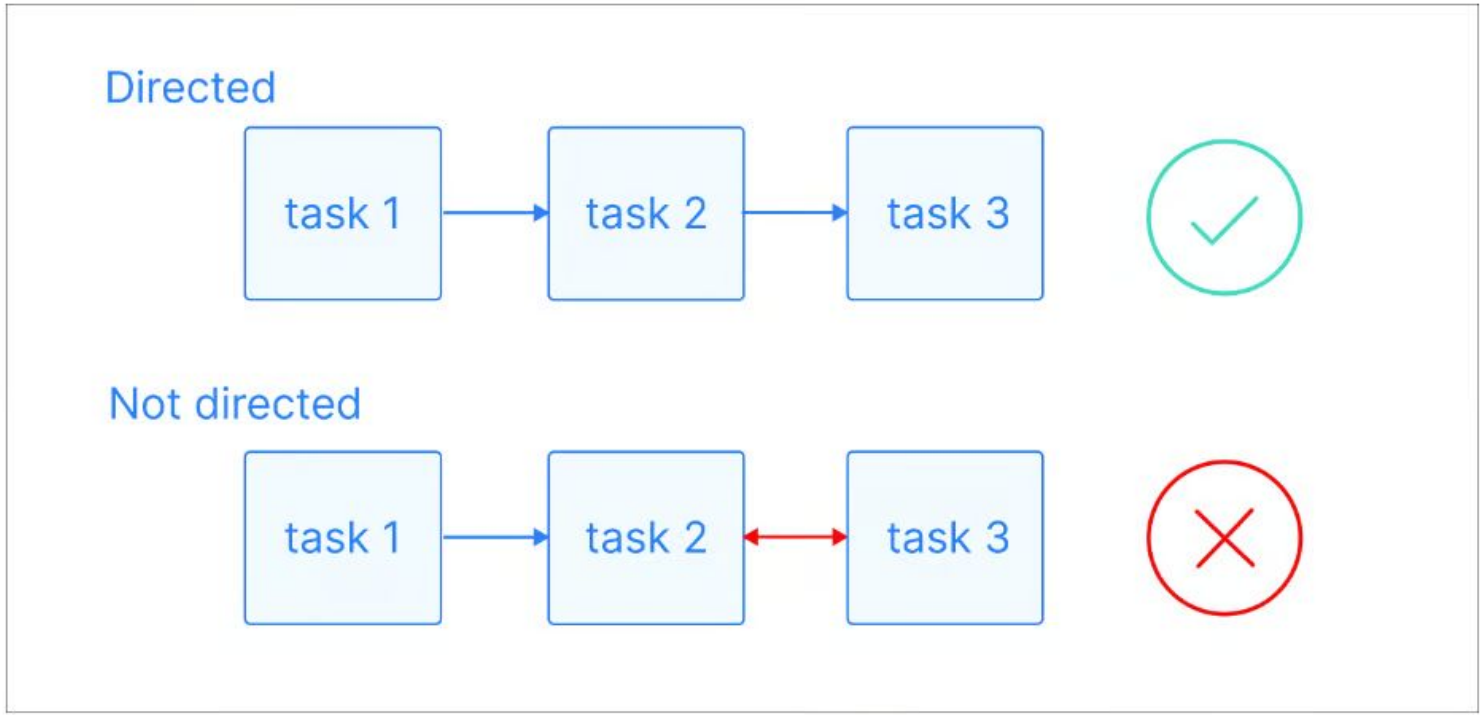
Componentes principales



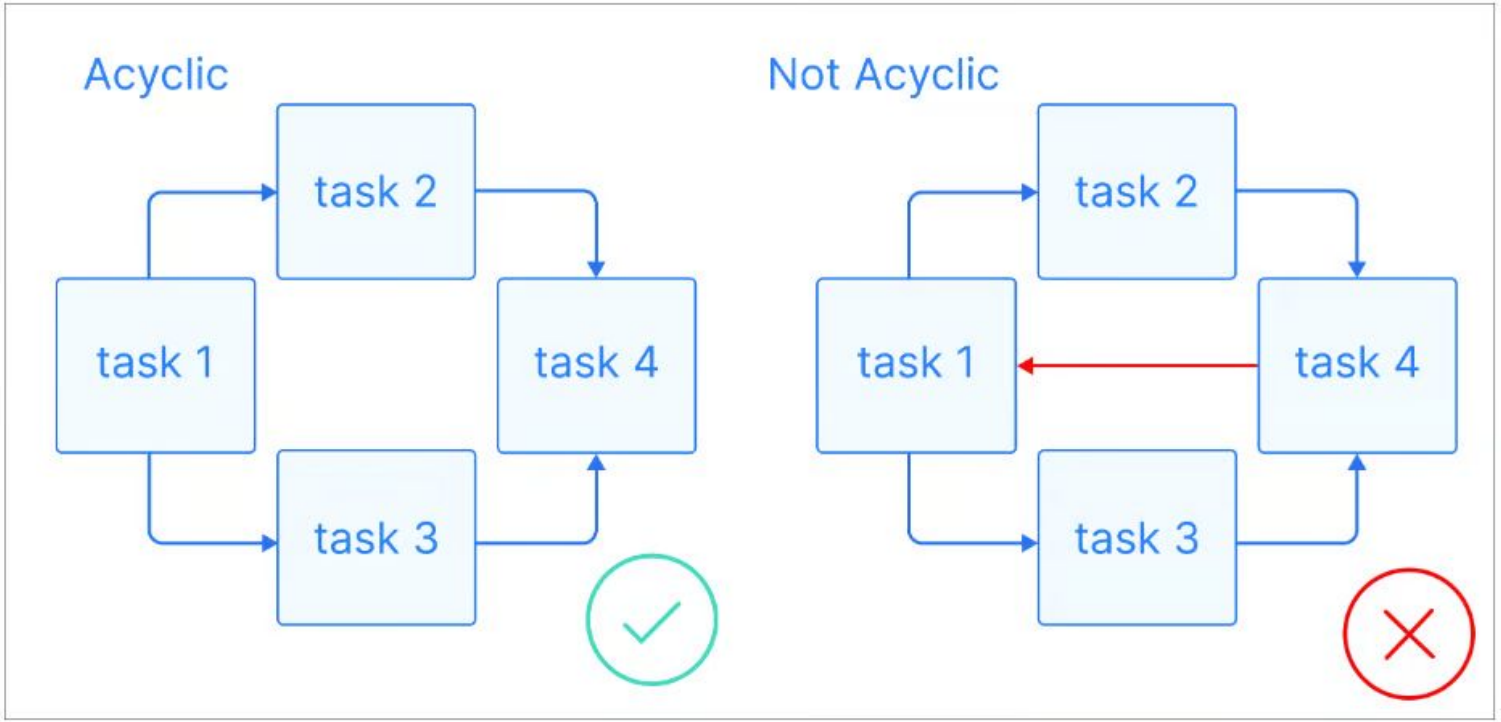
Componentes principales



¿Qué es un DAG? Directed Acyclic Graph



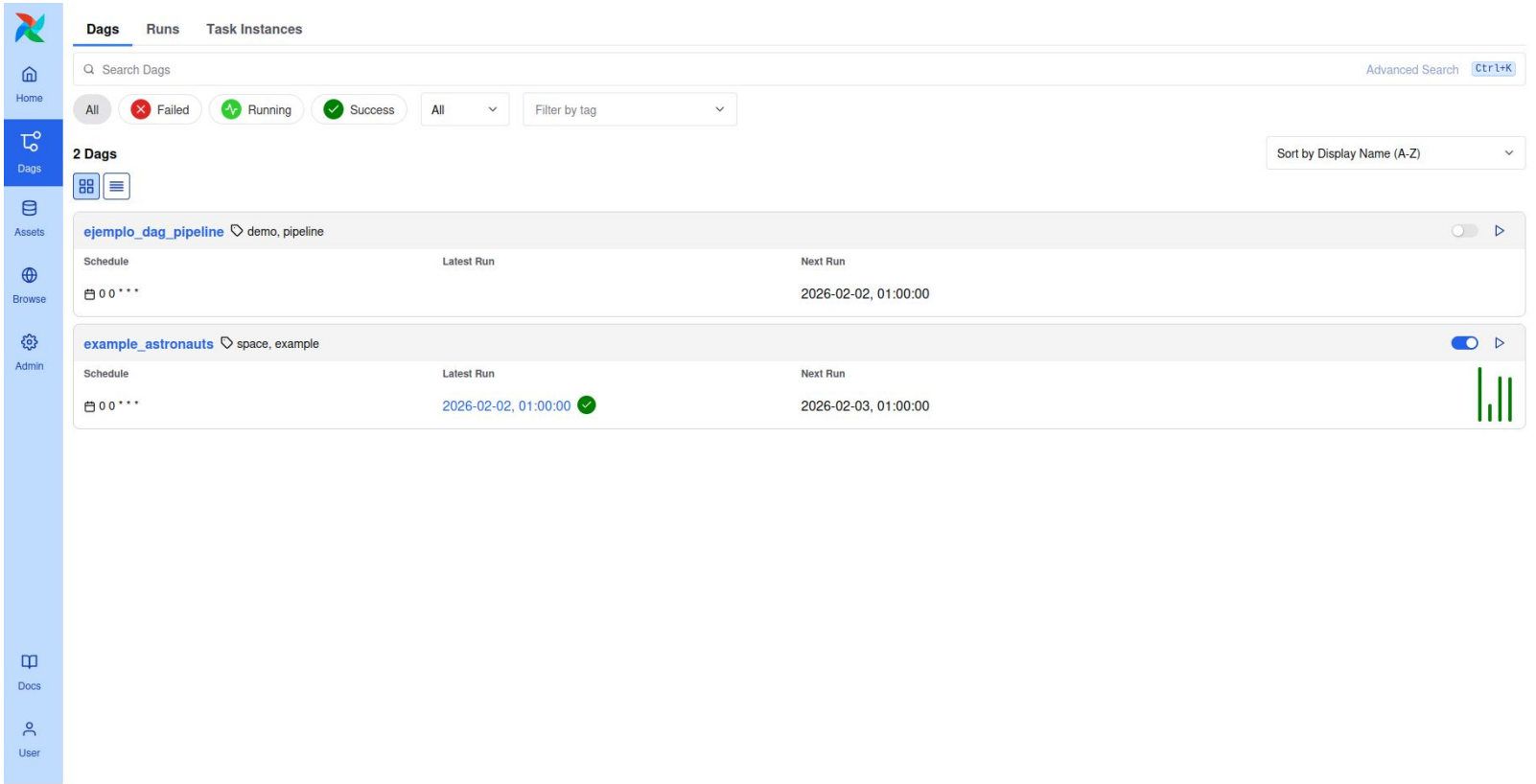
¿Qué es un DAG? Directed Acyclic Graph





04_
Primer vistazo

Primer vistazo - UI



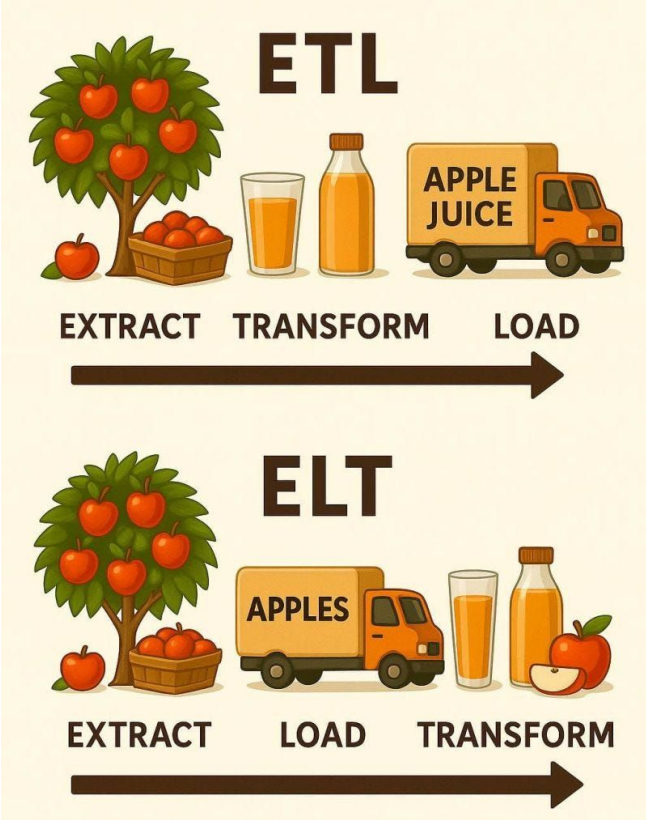
Primer vistazo - El dag



→

Workflow

Enfoque ETL vs ELT



Primer vistazo - Decoradores

```python
@dag(
    start_date=datetime(2025, 4, 1),
    schedule="@daily",
    max_consecutive_failed_dag_runs=5,
    doc_md=__doc__,
    default_args={
        "owner": "Astro",
        "retries": 3,
        "retry_delay": duration(seconds=5),
    },
    tags=["example", "space"],
    is_paused_upon_creation=False,
)
```

```python
@task_group(group_id="proceso_limpieza")
def seccion_limpieza():
    @task
    def borrar_nulos():
        return "Nulos borrados"

    @task
    def normalizar_fechas():
        return "Fechas corregidas"

    borrar_nulos() >> normalizar_fechas()

@task
def enviar_reporte():
    print("Proceso terminado")
```

Primer vistazo - Sin Decoradores

```python
### ETL PROCESS ###

def extract_data(ti=None, **kwargs):
    raw_data = "Raw data"
    ti.xcom_push(key="raw_data", value=raw_data)

def transform_data(ti=None, **kwargs):
    raw_data = ti.xcom_pull(task_ids="extract", key="raw_data")
    processed_data = f"Transformed data: {raw_data}"
    ti.xcom_push(key="processed_data", value=processed_data)

dag = DAG(
    dag_id="my_first_dag",
    start_date=datetime(2026, 1, 1),
    schedule_interval="30 20 * * *",
    catchup=False
)
```

DAGs anteriormente (sin decorators)

Primer vistazo - Operadores



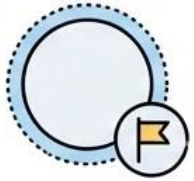
a. PythonOperator (ahora decorator @task)

Transformaciones ligeras, validaciones y lógica condicional.



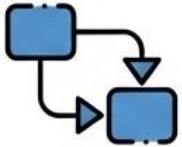
b. BashOperator

Ejecución de comandos o scripts.



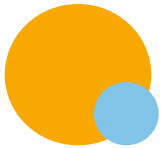
c. EmptyOperator (antes DummyOperator)

Creación de puntos de control, sin agregar funcionalidad.



d. >>

Define la dependencia entre dos tareas y controla el orden de ejecución.



Primer vistazo - Operadores



- Operadores más destacados de proveedores:
  - *SQL*

`BaseSQLOperator`	This is a base class for generic SQL Operator to get a DB Hook.
`SQLExecuteQueryOperator`	Executes SQL code in a specific database.
`SQLColumnCheckOperator`	Performs one or more of the templated checks in the column_checks dictionary.
`SQLTableCheckOperator`	Performs one or more of the checks provided in the checks dictionary.
`SQLCheckOperator`	Performs checks against a db.
`SQLValueCheckOperator`	Performs a simple value check using sql code.
`SQLIntervalCheckOperator`	Check that metrics given as SQL expressions are within tolerance of the ones from days_back before.
`SQLThresholdCheckOperator`	Performs a value check using sql code against a minimum threshold and a maximum threshold.
`BranchSQLOperator`	Allows a DAG to "branch" or follow a specified path based on the results of a SQL query.
`SQLInsertRowsOperator`	Insert rows (e.g. a collection of tuples) into a database table directly from an XCom or Python data structure.

Primer vistazo - Operadores

- Operadores más destacados de proveedores:
  - *Http*

`HttpOperator`	Calls an endpoint on an HTTP system to execute an action.

  - *Docker*

`DockerOperator`	Execute a command inside a docker container.

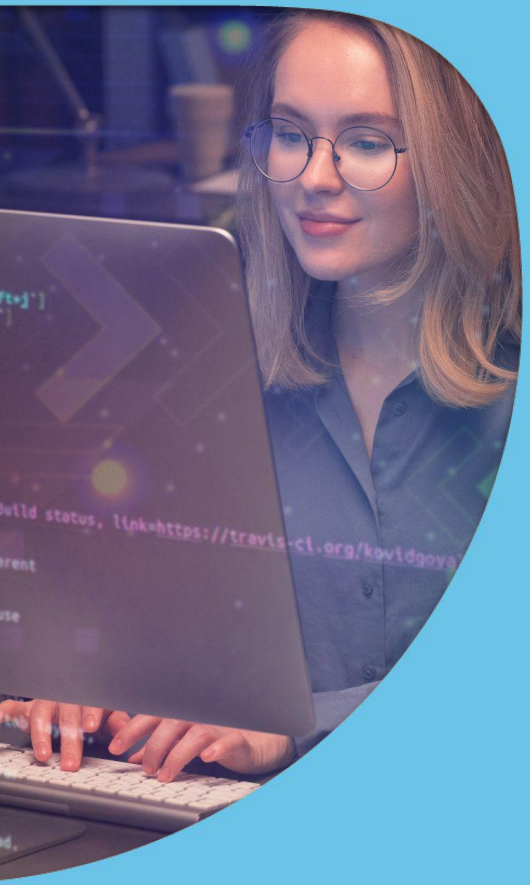
  - *Hive*

`HiveOperator`	Executes hql code or hive script in a specific Hive database.

Primer vistazo - Operadores

- Operadores más destacados de proveedores:
  - *BigQuery*

`BigQueryUIColors`	Hex colors for BigQue
`IfExistAction`	Action to take if the re
`BigQueryCheckOperator`	Performs checks agai
`BigQueryValueCheckOperator`	Perform a simple valu
`BigQueryIntervalCheckOperator`	Check that the values
`BigQueryColumnCheckOperator`	Subclasses the SQLCo
`BigQueryTableCheckOperator`	Subclasses the SQLTab
`BigQueryGetDataOperator`	Fetch data and return
`BigQueryCreateTableOperator`	Creates a new table in
`BigQueryDeleteDatasetOperator`	Delete an existing dataset from your Project in BigQuery.
`BigQueryDeleteDatasetOperator`	Delete an existing dataset from your Project in BigQuery.
`BigQueryCreateEmptyDatasetOperator`	Create a new dataset for your Project in BigQuery.
`BigQueryGetDatasetOperator`	Get the dataset specified by ID.
`BigQueryGetDatasetTablesOperator`	Retrieve the list of tables in the specified dataset.
`BigQueryUpdateTableOperator`	Update a table for your Project in BigQuery.
`BigQueryUpdateDatasetOperator`	Update a dataset for your Project in BigQuery.
`BigQueryDeleteTableOperator`	Delete a BigQuery table.
`BigQueryUpsertTableOperator`	Upsert to a BigQuery table.
`BigQueryUpdateTableSchemaOperator`	Update BigQuery Table Schema.
`BigQueryInsertJobOperator`	Execute a BigQuery job.

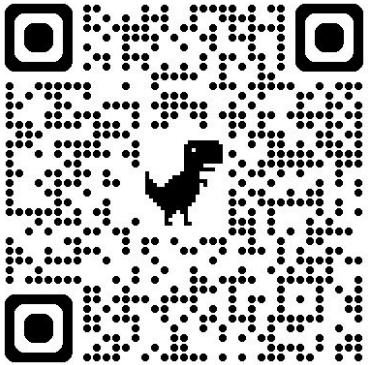


05_
Toma de contacto

Manos a la obra



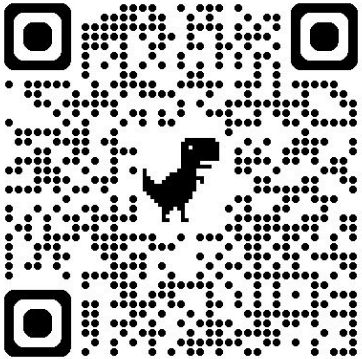
Toma de contacto



Homebrew



Instalación
Astronomer



Repositorio Github
con ejemplos

Gracias!

aulaupm@nextdigital.es

Orquesta tus datos, no tus dolores de cabeza

Anónimo

